

AttackCast: Predictive Cyber Attack Forecasting

AI-Powered Real-Time MITRE ATT&CK; World Model & Horizon Forecaster

1. Executive Summary & Problem Statement

Modern Security Operations Centers (SOCs) operate in a **reactive** state: analysts only investigate after an alert fires or damage is already done. **AttackCast** transforms defensive cybersecurity from *reactive detection* to **proactive, multi-step horizon forecasting**.

By analyzing streaming network packet flows in 30-second temporal windows, AttackCast's deep learning World Model (PyTorch GRU Neural Network) identifies the attacker's current kill-chain stage and forecasts where they will pivot **1 to 5 steps into the future**, recommending automated MITRE D3FEND countermeasures before critical impact occurs.

2. Key Innovations & Pillars

Pillar	What It Does	Impact on Security
Horizon Forecasting (\$K\$-step reachability)	Calculates future state probability distributions (\$k=1 \dots 5\$) using an empirical Markov transition matrix.	Gives defenders lead-time to pre-emptively neutralize attacks.
Explainable AI (XAI / SHAP)	Extracts top contributing flow features (Flow Bytes/s, PSH Flags, IAT Mean, etc.) with attribution scores.	Eliminates 'black box' AI; builds instant trust for SOC analysts.
MITRE D3FEND Countermeasures	Context-aware defense recommendations dynamically matched to the active kill-chain phase.	Automates actionable response plans (D3-IPC, D3-DQ, etc.).
What-If Counterfactual Sandbox	Interactive simulator allowing defenders to test hypothetical interventions (e.g. isolate host, block IP).	Proves how a defense instantly drops downstream compromise risk.

3. MITRE ATT&CK; 8-Stage Taxonomy

All streaming flows and real dataset samples are mapped to 8 standardized progression states:

Stage #	Stage Name	MITRE ID	Traffic Profile / Everyday Analogy
0	Benign / Normal	None	Regular daily network activity (safe baseline).
1	Reconnaissance	T1046	Port scanning, service discovery, high SYN flag rate.
2	Initial Access / Credential Access	T1110	SSH/RDP brute force, web vulnerability probing.
3	Execution / Foothold	T1059	Exploit payload delivery, reverse shell execution.
4	Command and Control (C2)	T1071	Periodic beaconing to external botnet controller IP.
5	Lateral Movement	T1021	Internal pivoting across hosts via SMB/WinRM/SSH.
6	Exfiltration	T1041 / T1048	Unusual spike in outbound bytes/sec to external IP.
7	Impact	T1498 / T1499	Critical denial of service (DDoS) or data encryption.

4. End-to-End System Architecture

AttackCast operates on a modern, high-throughput fullstack architecture consisting of three core layers:

Layer	Technology	Responsibilities
AI / ML Brain	PyTorch, Scikit-Learn, NumPy, Pandas	• 2-layer GRU Neural Network (Sequence modeling) • Ingestion of 2,520,751 real flows (CICIDS-2017) • Feature scaling (16 CICFlowMeter features) • Empirical Markov Transition Matrix computation
Backend API & Replay	FastAPI, Uvicorn, WebSockets, Python 3.11	• Real-time WebSocket packet stream server (<code>/ws/replay`</code>) • REST endpoints for health, metrics, what-if, countermeasures • PCAP file analysis and feature extraction engine
Frontend & Cockpit	React, TypeScript, TailwindCSS, Lucide, Vite	• Live attack monitor cockpit with threat level gauge • Interactive Host Interaction Network Graph • What-If Counterfactual Sandbox & Model Performance hub • Fully responsive mobile sidebar drawer

5. Real Dataset & Model Verification

AttackCast does not rely on random values. It is trained on real, empirical benchmark network data:

- **Benchmark Dataset:** CICIDS-2017 (717 MB raw CSV, 2,520,751 total network flows).
- **Extracted Features (16):** Destination Port, Flow Duration, Flow Bytes/s, Flow Packets/s, Fwd/Bwd Packet Length Means, Flow IAT Mean, Flag Counts (FIN, PSH, ACK), Initial Window Bytes, Active/Idle Means.
- **Trained PyTorch Model Checkpoint:** artifacts/models/world_model.pt (178 KB binary).
- **Empirical Transition Matrix:** Real probability distribution of attack phase shifts.

6. Platform UI Capabilities Walkthrough

Page / View	Key Analyst Features
Live Monitor (<code>/live`</code>)	Mission control showing live stage, dial alert levels (Green → Red), active flagged flows, top SHAP features, and dynamic D3FEND countermeasures.
Campaign / Host Graph (<code>/campaign`</code>)	Interactive network topology graph with live risk-weighted nodes and flow edges for single-host and multi-host coordinated attacks.
What-If Sandbox (<code>/whatif`</code>)	Defensive counterfactual simulator — test defensive interventions (e.g. isolate host, block IP) and see risk drop in real time.
Model Performance (<code>/performance`</code>)	Full AI validation hub: dataset card, confusion matrix, LR baseline vs GRU comparison, and training convergence curves.
PCAP Upload (<code>/pcap`</code>)	Drag-and-drop network capture ingestion hub for custom PCAP files.

7. Live Production Deployments & Access URLs

All services, cloud models, and interfaces are deployed and live on the public web:

Component / Service	Live URL	Environment / Host
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Frontend Web Cockpit	https://attack-cast-neon.vercel.app/	Vercel (Global Edge Network)
Cloud Backend API	https://attackcast.onrender.com/	Render Cloud (FastAPI + PyTorch)
Interactive Swagger Docs	https://attackcast.onrender.com/docs	OpenAPI Interactive Playground
Live API Health & Dataset Check	https://attackcast.onrender.com/api/health	Returns 2.52M flow dataset status
Live Model Metrics & Transition Matrix	https://attackcast.onrender.com/api/metrics	Returns F1 & transition matrix
GitHub Source Repository	https://github.com/Kumaran2048/AttackCast	GitHub (Full codebase + weights)